Local Implementation and Evaluation of Tiny Recursive Models (TRM) for Decision Support in Colombian SMEs Under Data and Compute Constraints

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Abstract

Small and medium-sized enterprises (SMEs) in Colombia represent 99.5% of formal businesses and generate roughly 80% of formal employment, yet they face severe barriers to adopting artificial intelligence (AI) due to limited historical data, modest hardware, and intermittent internet connectivity. Inspired by Jolicoeur-Martineau's Tiny Recursive Models (TRM), this study adapts and evaluates a lightweight recursive neural architecture for tabular business decision-making under strict local, offline constraints. Using a 1,000-sample credit risk dataset as an empirical proxy for SME records, we benchmark a 16,641-parameter Tabular TRM against a Standard 2-Layer MLP and XGBoost on a standard CPU. Results show that XGBoost achieves the highest accuracy (0.7350) and F1-score (0.8166) in 0.10s, while Tabular TRM achieves comparable performance (0.6750 accuracy, 0.7653 F1-score) to the Standard MLP (0.6800 accuracy, 0.7681 F1-score) with an offline inference latency of 0.0109 ms/sample. These findings demonstrate that while tiny recursive networks can operate locally without cloud dependencies, tree-based ensembles remain more sample-efficient for small tabular SME datasets.

1. Introduction

In Colombia, micro, small, and medium-sized enterprises (MSMEs) account for 99.5% of active formal companies and approximately 80% of formal employment. Despite their economic weight, critical operational decisions—such as credit risk detection, inventory management, and demand forecasting—are still predominantly handled using spreadsheets or basic statistical rules.

While deep learning and Large Language Models (LLMs) excel at complex reasoning, they typically require massive datasets, specialized GPUs, and continuous cloud connectivity. These requirements are incompatible with many Colombian SMEs, particularly in regions with intermittent internet or limited IT infrastructure. Recently, Jolicoeur-Martineau (2025) introduced Tiny Recursive Models (TRM), demonstrating that a single 2-layer network (~7M parameters) iterating recursively over a latent state can outperform massive LLMs on structured reasoning benchmarks like ARC-AGI and Sudoku-Extreme using ~1,000 training samples. However, no prior work has evaluated TRM on real-world tabular business tasks under SME hardware constraints. This paper bridges that gap by implementing and benchmarking a Tabular TRM locally on CPU.

2. Related Work

- **Recursive Reasoning Models:** Hierarchical Reasoning Models (HRM) introduced multi-timescale recursion with deep supervision. TRM simplified this paradigm by eliminating fixed-point theorems and using a single 2-layer network that iteratively refines an output y and latent state z given input x, achieving 87.4% accuracy on Sudoku-Extreme.
- **Small-Sample Learning in SMEs:** Prior studies on efficient machine learning for SMEs emphasize that data scarcity, noise, and lack of dedicated hardware make standard deep learning infeasible, favoring compact, interpretable, and locally deployable models.

3. Methodology

Following an adapted CRISP-DM workflow, we evaluate three models on a **Credit Risk Detection** task using the German Credit dataset (credit-g, $N = 1{,}000$ samples, 20 raw features expanded to 61 standardized/one-hot encoded features) with an 80/20 stratified train-test split:

1. **XGBoost (Traditional ML Baseline):** 100 estimators, max depth 4, learning rate 0.1.
2. **Standard MLP (Deep Learning Baseline):** Two hidden layers (64 units each), ReLU activations, dropout (0.2), totaling **8,193 parameters**.
3. **Tabular TRM (Proposed):** An input projection to $d_{\text{latent}} = 64$, followed by a single 2-layer recursive block (LayerNorm + GELU) iterating $T = 4$ steps with residual latent updates ($z_{t+1} = z_t + f(x, z_t, \hat{y}_t)$) and deep supervision across all steps, totaling **16,641 parameters**.

All neural models were trained for 40 epochs using AdamW (lr $= 10^{-3}$) on a standard **CPU** (no GPU acceleration) to simulate realistic SME hardware.

4. Experimental Results

Table 1 summarizes the predictive performance, training time, and offline inference latency across all three models.

Model	Accuracy	F1-Score	Training Time (CPU)	Inference Latency	Parameters
XGBoost	**0.7350**	**0.8166**	**0.10 s**	0.0165 ms/sample	—

Model	Accuracy	F1-Score	Training Time (CPU)	Inference Latency	Parameters
Standard MLP	0.6800	0.7681	2.22 s	**0.0029 ms/sample**	8,193
Tabular TRM ($T = 4$)	0.6750	0.7653	5.05 s	0.0109 ms/sample	16,641

Key Observations:

- **Predictive Performance:** XGBoost outperformed both neural architectures by +5.5% in accuracy and ~+5% in F1-score, confirming the strong inductive bias of gradient-boosted trees on small, structured tabular datasets.
- **TRM vs. Standard MLP:** Tabular TRM achieved virtually identical accuracy (0.6750 vs. 0.6800) and F1-score (0.7653 vs. 0.7681) to the Standard MLP, indicating that recursive latent refinement did not yield an accuracy advantage over a standard feedforward pass on 1,000 tabular rows.
- **Local Offline Feasibility:** All three models trained in **under 5.1 seconds** on a standard CPU and achieved **sub-millisecond inference** (<0.02 ms/sample), proving complete operational autonomy without cloud APIs or GPUs.

5. Conclusion and Future Work

This study evaluated the feasibility of adapting Tiny Recursive Models (TRM) for offline, local decision support in Colombian SMEs. While Tabular TRM proved lightweight enough to train and run on a standard CPU in seconds, XGBoost remains the superior choice for small tabular datasets (~1,000 samples) in both accuracy and training speed. Future work should explore whether TRM's recursive reasoning provides measurable gains on **sequential or multi-step SME tasks** (such as multi-echelon inventory optimization or route planning) rather than static tabular classification.

References

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